How the Tax Report PDF Is Generated

A tool that computes capital gains on E*TRADE / Revolut shares under Spanish tax rules

Presentation for the tax advisor / accountant

> ⚠️ The tool is an aid. **The final figures are always validated by you.**

The idea in one sentence

> We take the **raw** data from the broker (purchases, vestings, exercises and sales), apply the **FIFO** method and the **Spanish IRPF** rules, and pour the result into a **PDF** organised by **Modelo 100** boxes.

One data source → one calculation → one report.

Nothing is entered by hand.

The data journey (4 steps)

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1. RAW DATA                 2. TAX ENGINE             3. TEMPLATE            4. PDF
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E*TRADE (Excel):    →       FIFO + IRPF      →        HTML report     →      Chromium
 · ESPP purchases           rules:                    with all the            prints
 · RSU vestings              · FIFO ordering          sections and            the HTML
 · Option exercises          · ESPP art. 42.3.f       tables                  to A4
 · Sales                     · 2-month rule                                   (final PDF)
Revolut (CSV):               · savings base
 · Trades                    · dividends/interest
 · Dividends
```

Every figure in the PDF can be traced back to a specific row in the broker's Excel.

Step 1 — Where the data comes from

It is read **directly from the files E*TRADE exports** (and, if present, the Revolut CSV):

Event type	Source	What it provides
ESPP purchases	E*TRADE Excel	Purchase price and company discount
RSU vestings	E*TRADE Excel	Vested shares and their value
Option exercises	E*TRADE Excel	Acquisition cost
Sales	E*TRADE Excel	Date, number of shares and sale price
Revolut (optional)	CSV	Trades and dividends for the same security

No figures are entered by hand: the official broker files are parsed.

Step 2 — The tax engine (the heart of the calculation)

Every event passes through **a single engine** that applies, in this order:

1. **FIFO** — the shares sold are the **oldest** ones first (mandatory criterion in Spain; specific-lot identification is not allowed).
2. **EUR conversion** — each operation is converted using the **ECB exchange rate** for its date (not an annual average rate).
3. **Automatic *sell-to-cover* detection** — sales used to cover withholding at vesting are identified and classified automatically.
4. **Special rules** (next slide).

Step 2 — Tax rules it applies

Rule	Legal basis	What it does
FIFO method	Art. 37.2 LIRPF	Matches sales with the oldest purchases
ESPP exemption	Art. 42.3.f LIRPF	The company discount is exempt if shares are held for **3 years**; if sold earlier, it becomes **employment income**
2-month rule	Art. 33.5.f LIRPF	Blocks losses if the same security is rebought within ±2 months (anti *wash-sale*)
Savings base	Art. 49 LIRPF	Integrates gains + dividends + interest and offsets losses
4-year carryforward	Art. 49 LIRPF	Carries pending losses to future years

Steps 3 and 4 — From calculation to PDF

- The engine's result fills an **HTML report template** with all the tables.
- An (automated) **Chromium** browser **prints that HTML to PDF** in A4 format.
- **Two versions** are produced: 🇪🇸 Spanish and 🇺🇸 English, with the **same numbers**.

> The PDF is, literally, the engine's calculation "printed". It is not edited afterwards.

What the PDF contains (section by section)

1. **Methodology** — explains FIFO and the rules applied.
2. **Portfolio summary by security** — remaining shares and average cost.
3. **Yearly Tax Summary** — gain/loss per year (Savings Base, Modelo 100).
4. **Gains/losses by broker** — E*TRADE vs Revolut breakdown.
5. **Savings Base** — gains + dividends/interest (Art. 49).
6. **Loss carryforward ledger** — 4-year carryover.

What the PDF contains (continued)

7. **Disposals detail** — each sale, ready for the Modelo 100.
8. **Modelo 100 filing guide** — which box to fill with which figure.
9. **ESPP 3-year analysis** — exempt vs taxable (Art. 42.3.f).
10. **Detailed transaction ledger** — each operation with its step-by-step FIFO calculation.

> Sections 7 and 8 are the ones designed **directly for the tax return**.

Example 1 — A sale with FIFO step by step

Setup. You hold two RSU lots of the same security and sell part of them:

Lot	Date (vesting)	Shares	Price	ECB rate	Cost in EUR
Lot 1	15/03/2022	10	$100	0.90	**€900.00**
Lot 2	20/06/2023	10	$120	0.92	**€1,104.00**

➡️ **You sell 15 shares** on 10/09/2024 at $150, ECB rate 0.91.

Example 1 — The engine's calculation

Sale proceeds (EUR): 15 × \$150 × 0.91 = **€2,047.50**

FIFO → oldest shares are sold first: 10 from Lot 1 + 5 from Lot 2

Shares	From	Cost EUR
10	Lot 1 (all)	€900.00
5	Lot 2 (5 of 10 → 5×120×0.92)	€552.00
15		**€1,452.00**

Capital gain = 2,047.50 – 1,452.00 = **€595.50** → Savings Base

Remaining in portfolio: 5 shares from Lot 2 (cost €552.00), ready for the next sale.

Example 2 — The ESPP discount (Art. 42.3.f)

Setup. ESPP purchase: market price $100, you pay 85% = $85.
Company discount = **$15/share × 10 shares = $150**.

Scenario	Tax treatment
🟢 You hold the shares **≥ 3 years**	The $150 discount is **EXEMPT** (not taxed)
🔴 You sell **before 3 years**	The $150 (converted to EUR) becomes **employment income** and is taxed

> The report places each case in its section: exempt → informational only; early sale → employment income. **The gain/loss on the share sale is computed separately, always via FIFO.**

Example 3 — The Savings Base (Art. 49)

Gains, dividends and interest are integrated, and losses are offset:

Item (year 2024)	Amount
Capital gain (Example 1)	+€595.50
Dividends / interest (converted to EUR)	+€200.00
Losses from other sales	–€100.00
Net savings base	**€695.50**

Progressive savings scale: 19% up to €6,000 · 21% from €6,000–50,000 · 23% from €50,000–200,000 · 27% from €200,000–300,000 · 28% above.

➡️ Here: €695.50 × 19% ≈ **€132.15 estimated tax**.

Example 4 — The 2-month rule (anti *wash-sale*)

Setup. You sell at a **loss** and rebuy the same security shortly after:

Operation	Date	Result
Sale at a −€300 loss	10/05/2024	Loss...
Rebuy of the **same security**	02/06/2024 (< 2 months)	🚫 Loss **blocked**

➡️ The −€300 loss **cannot be claimed now**: it is "parked" and added to the cost of the new shares (you will use it when you sell those for good).

> The report flags these losses as blocked so they are not declared by mistake.

How the examples appear in the PDF

Example	Report section where it appears
1 — FIFO sale	*Disposals detail* + *Detailed transaction ledger*
2 — ESPP	*ESPP 3-year analysis* (exempt) or *Employment income* (early sale)
3 — Savings base	*Yearly Tax Summary* + *Savings Base*
4 — 2-month rule	*Loss carryforward ledger* (blocked loss)

All amounts in EUR, using the **ECB rate for each date**.

Why every number can be trusted

- **Full traceability**: each report row comes from a row in the broker's Excel.
- **Official exchange rate**: ECB, by operation date (with an auditable cache).
- **No manual entry**: the error of transcribing figures by hand is eliminated.
- **Same engine, two outputs**: the PDF (tax return) and the interactive dashboard (decision-making) share the same calculations; the anchor numbers must match.
- **Automated test coverage** over the tax rules (FIFO, ESPP, sale classification).

What the report does not do (limits)

- It does not file the return: it **produces the figures**, you enter/validate them.
- It does not replace your professional judgement or tax advice.
- It reflects **our interpretation** of the rules; you have the final word.
- "Live" prices (today's value) live in the dashboard, **not** in the PDF; the PDF is a **fixed tax snapshot**.

Summary for the advisor

> **Official broker data → FIFO method + IRPF rules → PDF organised by Modelo 100 boxes.**

- One source, one calculation, no manual edits.
- Every figure is **traceable** and converted to EUR using the **ECB rate** by date.
- The PDF separates what goes to **capital gains**, **employment income** (ESPP sold before 3 years) and **savings base** (dividends/interest).

What it needs from you? Review the *Disposals* and *Modelo 100 guide* sections and confirm the approach works for you.